Supporting Information
Semiempirical Orbital-Optimized DFT in MOUSE: Spin-Pure Excited-State Surfaces with Analytic Gradients and Hessians

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1 Data scope and fixed protocol

The data in this document are generated from the final relative-active-pair qualification archive, not from the earlier shared-pair or shared-hole exploratory surfaces. The two parent weights are 0.5, the exchange-response scaling is 0.9, and the additive excitation-energy offset is zero. The full and qualified-spectral parents share their quadratic charge/spin correction and differ in their one-body operator. The mixed-spin and triplet energies within each parent use common spatial orbitals.

The 15 regression states were exposed during development. The four final prospective molecules had previous native/shell-model predictions, but the present full/spectral composite protocol was frozen before their new predictions. Reference transition character, rather than TBE proximity, determines assignments. The reported states constitute a curated qualified cohort; the table is not a survey of every attempted state in preceding development campaigns. In particular, its size must not be used as a database-wide success-rate denominator.

The spectral matching uses occupation-sector assignments with absolute overlap at least 0.7 and a best-versus-alternative margin of at least 0.05. Unqualified orbital levels retain native spacings. There is no fitted orbital-extent correction and no multiplication of the orbital gaps by 0.9. The effective exact-exchange response coefficient is 0.225, and the semilocal response is unchanged. The mean occupied energy sets a gauge for spectral shifts, not a fitted shift of excitation energies.

The two relative active-pair targets are squared cross-parent overlaps at the anchor. They remain fixed while the absolute orbital orientations and spectators relax. Targets within 10^{-10} of unity use an exact shared-orbital elimination. The final stationarity certificate concerns the coupled Lagrangian; independent parent orbital gradients are not assumed to vanish away from the anchor. Projected absolute orbital curvatures must exceed $10^{-7}\ E_{\mathrm{h}}$ for certification. The full-CSF and active-character following thresholds are 0.7.

2 Complete vertical-energy results

Table S1: All reported vertical excitation energies and signed errors in eV. The final four rows are prospective tests; the preceding rows are exposed regression examples.

Molecule	State	TBE	OO-DFT	Error
Acetaldehyde	$n \to \pi^*$	4.304	4.277	-0.027
Acetone	$n \to \pi^*$	4.468	4.338	-0.130
Acrolein	$n \to \pi^*$	3.723	3.771	+0.048
Cyclopentadienone	$n \to \pi^*$	2.951	2.953	+0.002
Formamide	$n \to \pi^*$	5.644	5.648	+0.004
Glyoxal	$n \to \pi^*$	2.877	2.752	-0.125
Maleimide	$n \to \pi^*$	3.810	3.643	-0.167
Nitrosomethane	$n \to \pi^*$	1.952	1.721	-0.231
Nitroxyl	$n \to \pi^*$	1.743	1.485	-0.258
Pyranone	$n \to \pi^*$	4.016	4.076	+0.060
Pyrazine	$n \to \pi^*$	4.144	3.932	-0.212
Pyrimidine	$n \to \pi^*$	4.451	4.218	-0.233
Tetrazine	$n \to \pi^*$	2.462	2.060	-0.402
Uracil	$n \to \pi^*$	4.935	4.885	-0.050
Uracil	$\pi \to \pi^*$	5.381	5.686	+0.305
Cyanoformaldehyde	$n \to \pi^*$	3.817	3.989	+0.172
Cyclopropenone	$n \to \pi^*$	4.274	4.482	+0.208
Imidazole	$n \to \pi^*$	6.715	6.180	-0.535
Pyridazine	$n \to \pi^*$	3.830	3.530	-0.300

3 Direct dedicated sTDA-xTB comparison

The published dedicated Hamiltonian and parameters are retained. Ground-state xtb4stda wavefunctions and std2 singlet vectors are evaluated at the original XYZs, without geometry optimization or TBE-based root assignment. Cross-basis transition overlaps are normalized only after projection; the norm, overlap, and assignment margin are reported separately in `data/direct_stda_comparison.csv`. All 19 states pass overlap ≥ 0.7, projected norm ≥ 0.7, and margin ≥ 0.05. The small difference between coordinate conversions stored by the legacy executable and the input XYZ is bounded by 2×10^{-6} bohr; no geometry is optimized or refitted. Earlier parameter-path setup failures are preserved separately from the successful electronic attempts.

Table S2: Every direct sTDA-xTB comparison. Energies and signed errors are in eV; q is the absolute normalized cross-AO transition overlap.

Molecule	State	TBE	MOUSE	sTDA-xTB	Δ_{sTDA}	q
Acetaldehyde	$n\pi^*$	4.304	4.277	4.454	+0.150	0.915
Acetone	$n\pi^*$	4.468	4.338	4.531	+0.063	0.930
Acrolein	$n\pi^*$	3.723	3.771	3.826	+0.103	0.918
Cyanoformaldehyde	$n\pi^*$	3.817	3.989	3.911	+0.094	0.900
Cyclopentadienone	$n\pi^*$	2.951	2.953	2.840	-0.111	0.923
Cyclopropenone	$n\pi^*$	4.274	4.482	3.838	-0.436	0.902
Formamide	$n\pi^*$	5.644	5.648	6.039	+0.395	0.853
Glyoxal	$n\pi^*$	2.877	2.752	2.007	-0.870	0.924

Molecule	State	TBE	MOUSE	sTDA-xTB	Δ_{sTDA}	q
Imidazole	$n\pi^*$	6.715	6.180	7.277	+0.562	0.739
Maleimide	$n\pi^*$	3.810	3.643	3.654	-0.156	0.931
Nitrosomethane	$n\pi^*$	1.952	1.721	0.492	-1.460	0.950
Nitroxyl	$n\pi^*$	1.743	1.485	0.237	-1.506	0.955
Pyranone	$n\pi^*$	4.016	4.076	4.710	+0.694	0.913
Pyrazine	$n\pi^*$	4.144	3.932	4.213	+0.069	0.910
Pyridazine	$n\pi^*$	3.830	3.530	3.810	-0.020	0.921
Pyrimidine	$n\pi^*$	4.451	4.218	4.684	+0.233	0.932
Tetrazine	$n\pi^*$	2.462	2.060	1.753	-0.709	0.946
Uracil	$n\pi^*$	4.935	4.885	5.760	+0.825	0.892
Uracil	$\pi\pi^*$	5.381	5.686	5.210	-0.171	0.884

4 Physical structural accuracy and weight precheck

The following internal-coordinate references are from the published CCSDR(3)/def2-TZVPP supporting tables (DOI 10.1021/acs.jctc.7b00921): acetaldehyde S19, acetone S20, and maleimide S27. The geometry definitions and reference provenance are retained in `data/geometry_reference_comparison.json`. No structural reference was used to fit the energy model.

Table S3: Internal coordinates of MOUSE excited-state minima against CCSDR(3)/def2-TZVPP. These comparisons assess physical accuracy separately from derivative correctness.

Molecule	Coordinate	MOUSE	Reference	Error	Unit
Acetaldehyde	C-O	1.312	1.331	-0.019	Å
Acetaldehyde	C-C	1.488	1.508	-0.020	Å
Acetaldehyde	C-Hald	1.098	1.091	+0.007	Å
Acetaldehyde	C-C-O	116.841	115.200	+1.641	°
Acetaldehyde	C-C-Hald	113.767	119.300	-5.533	°
Acetaldehyde	pyramidalization	41.651	38.700	+2.951	°
Acetone	C-O	1.319	1.336	-0.017	Å
Acetone	C-C	1.489	1.507	-0.018	Å
Acetone	C-C-C	117.761	119.500	-1.739	°
Acetone	pyramidalization	30.555	42.300	-11.745	°
Maleimide	N-C	1.378	1.404	-0.026	Å
Maleimide	C-O	1.244	1.248	-0.004	Å
Maleimide	C-C	1.426	1.432	-0.006	Å
Maleimide	C=C	1.371	1.386	-0.015	Å
Maleimide	N-H	1.012	1.006	+0.006	Å
Maleimide	C-H	1.072	1.074	-0.002	Å
Maleimide	C-N-C	111.090	109.800	+1.290	°

The weight table uses independently stationary vertical parents only. It cannot substitute for reoptimizing a weighted off-anchor coupled problem. The production weight remains 0.5. The separate harmonic nitroxyl precheck is retained in `data/hno_harmonic_comparison.json`; its frequency-order correspondence remains provisional until mode-vector confirmation.

Table S4: Algebraic weight sensitivity for the 19 independently stationary vertical-anchor parent gaps. This is not a test of the weighted off-anchor potential or its Hessian. The production weights remain 0.5/0.5.

w_F	MAE (eV)	RMSE (eV)	Max. error (eV)
0.00	0.210	0.257	0.645
0.25	0.179	0.211	0.590
0.50	0.183	0.229	0.535
0.75	0.239	0.300	0.583
1.00	0.308	0.396	0.958

5 Reference data provenance

The machine-readable file `data/tbe_provenance.json` associates each state with its archived QUEST TBE method. The target basis is aug-cc-pVTZ (AVTZ); AVDZ denotes aug-cc-pVDZ. These are theoretical vertical references, not measured absorption maxima. The full input metadata and inherited state assignments are in `reference_provenance.json`. The DFT reference basis is aug-cc-pVDZ; it is distinct from the target basis of the TBE. These values are used for reporting errors, not constructing the semiempirical energy.
Acetaldehyde (n_pi): CCSDTQ/AVDZ + [CCSDT/AVTZ - CCSDT/AVDZ].
Acetone (n_pi): CCSDTQ/6-31+G(d) + [CC4/AVDZ - CC4/6-31+G(d)] + [CCSDT/AVTZ - CCSDT/AVDZ].
Acrolein (n_pi): CCSDTQ/6-31+G(d) + [CC4/AVDZ - CC4/6-31+G(d)] + [CCSDT/AVTZ - CCSDT/AVDZ].
Cyclopentadienone (n_pi): CC4/6-31+G(d) + [CCSDT/AVTZ - CCSDT/6-31+G(d)].
Formamide (n_pi): CCSDTQ/AVDZ + [CCSDT/AVTZ - CCSDT/AVDZ].
Glyoxal (n_pi): CC4/AVDZ + [CCSDT/AVTZ - CCSDT/AVDZ].
Maleimide (n_pi): CC4/6-31+G(d) + [CCSDT/AVTZ - CCSDT/AVDZ].
Nitrosomethane (n_pi): CCSDTQ/AVDZ + [CCSDT/AVTZ - CCSDT/AVDZ].
Nitroxyl (n_pi): CCSDTQP/6-31+G(d) + [CCSDTQ/AVTZ - CCSDTQ/6-31+G(d)].
Pyranone (n_pi): CC4/6-31+G(d) + [CCSDT/AVTZ - CCSDT/6-31+G(d)].
Pyrazine (n_pi): CC4/6-31+G(d) + [CCSDT/AVTZ - CCSDT/6-31+G(d)].
Pyrimidine (n_pi): CC4/6-31+G(d) + [CCSDT/AVTZ - CCSDT/6-31+G(d)].
Tetrazine (n_pi): CC4/AVDZ + [CCSDT/AVTZ - CCSDT/AVDZ].
Uracil (n_pi): CCSDT/6-31+G(d) + [CC3/AVTZ - CC3/6-31+G(d)].
Uracil (pi_pi): CCSDT/6-31+G(d) + [CC3/AVTZ - CC3/6-31+G(d)].
Cyanoformaldehyde (n_pi): CC4/AVDZ + [CCSDT/AVTZ - CCSDT/AVDZ].
Cyclopropenone (n_pi): CC4/AVDZ + [CCSDT/AVTZ - CCSDT/AVDZ].
Imidazole (n_pi): CC4/6-31+G(d) + [CCSDT/AVTZ - CCSDT/6-31+G(d)].
Pyridazine (n_pi): CC4/6-31+G(d) + [CCSDT/AVTZ - CCSDT/6-31+G(d)].

6 Derivative checks and stationarity criteria

The five nuclear minima satisfy a maximum Cartesian gradient below 3×10^{-5} E_h/bohr and have positive projected vibrational curvatures. Positive excitation gaps and tracked state character are required as separate checks. Directional checks use central differences of the scalar energy and analytic gradient at $\pm hv$, with $h = 2 \times 10^{-4}$ and 10^{-4} bohr. The reported Hessian-vector discrepancy is the largest component of the difference from the analytic Hessian acting on the same direction. No finite differences enter the production gradient or Hessian.

At $h = 10^{-4}$ bohr, complete Cartesian-column checks yield:

Molecule	Columns	Maximum Hessian error	Scalar-gradient error
Nitroxyl	9	1.1512×10^{-8}	3.6434×10^{-9}
Formamide	18	4.6669×10^{-9}	2.2575×10^{-9}

Hessian errors are in $E_\mathrm{h}/\mathrm{bohr}^2$; scalar-gradient errors are in $E_\mathrm{h}/\mathrm{bohr}$. The maximum asymmetry of the numerical Hessians is 8.74×10^{-9} and 5.30×10^{-9} $E_\mathrm{h}/\mathrm{bohr}^2$, respectively. Full displacement checks and the two-step directional checks for every minimum are recorded in `data/SUMMARY.json`. Translational and rotational modes are removed before vibrational stability is assessed. No experimental-frequency accuracy is inferred from these implementation tests.

7 Optimized geometries and projected harmonic frequencies

All coordinates below are in Angstrom, in the original atom order. They describe minima of the constrained composite semiempirical surface. The XYZ files are also supplied in `coordinates/`. These coordinates are not high-level reference geometries.

7.1 Acetaldehyde

```
7
acetaldehyde; constrained OO-DFT minimum; coordinates in Angstrom
C   0.2223035093  0.1013277113  0.2334913667
C  -0.9453005961 -0.0251942583 -0.6804940117
O   0.4079243109 -0.8702860896  1.0954507793
H   0.3010447606  1.0841255171  0.7176125575
H  -1.8688390394 -0.3135713164 -0.1481562051
H  -0.7506835894 -0.8350717205 -1.3936582758
H  -1.1076978659  0.9035664200 -1.2182916609
```

Projected harmonic frequencies (cm^{-1}): 186.89, 432.23, 667.85, 973.26, 990.28, 1095.13, 1240.79, 1375.25, 1389.50, 1416.35, 1455.67, 2766.35, 2838.16, 2919.28, 3059.54.

7.2 Acetone

```
10
acetone; constrained OO-DFT minimum; coordinates in Angstrom
C  -0.5689937111 -0.0533836291 -0.2955897103
C   0.0475047310  1.2845796508 -0.5101289622
C   0.2806363777 -1.2536119814 -0.5275519803
O  -1.8450965398 -0.1685475055 -0.6076164069
H   1.0922037630  1.2729010514 -0.2125706389
H   1.3076035812 -1.0537316303 -0.2352636408
H  -0.0227253198  1.6074367061 -1.5645009174
H  -0.4976868798  2.0329363229  0.0742268604
H   0.2629093176 -1.5736462119 -1.5849234343
H  -0.1140660180 -2.0949327727  0.0514752619
```

Projected harmonic frequencies (cm^{-1}): 148.28, 149.63, 270.71, 368.65, 500.42, 852.57, 968.79, 988.72, 1000.93, 1047.16, 1229.45, 1371.00, 1387.39, 1394.90, 1416.89, 1425.45, 1460.92, 1468.31, 2741.86, 2761.53, 2932.39, 2934.40, 3043.80, 3048.12.

7.3 Formamide

```
6
formamide; constrained OO-DFT minimum; coordinates in Angstrom
C  -0.1991028844 -0.2725711755  0.5163934339
O   1.0835179824 -0.3792440676  0.1254002010
N  -1.0286807403  0.2161588604 -0.4725654836
H  -0.1480202140  0.3692094734  1.4479826708
H  -0.5794619260  0.4820208966 -1.3402912231
H  -1.8635741275 -0.3394673884 -0.6301074391
```

Projected harmonic frequencies (cm^{-1}): 324.38, 489.56, 567.75, 902.41, 1043.32, 1172.31, 1242.56, 1256.85, 1491.02, 2377.42, 3308.98, 3430.06.

7.4 Maleimide

```
10
maleimide; constrained OO-DFT minimum; coordinates in Angstrom
C   0.0000000000  1.1364919551  0.0927366553
C   0.0000000000 -1.1364919551  0.0927366554
C  -0.0000000000  0.6854283480 -1.2605427879
C   0.0000000000 -0.6854283480 -1.2605427880
N  -0.0000000000  0.0000000000  0.8725164373
O   0.0000000000  2.2842746107  0.5713837699
O   0.0000000000 -2.2842746107  0.5713837699
H   0.0000000000  0.0000000000  1.8848555803
H  -0.0000000000  1.3571912082 -2.0958657911
H  -0.0000000000 -1.3571912082 -2.0958657910
```

Projected harmonic frequencies (cm^{-1}): 179.17, 288.09, 385.45, 502.10, 527.27, 554.29, 601.09, 618.76, 648.38, 676.37, 705.14, 774.17, 992.59, 1029.27, 1049.28, 1165.93, 1220.57, 1252.23, 1312.02, 1486.37, 1636.37, 3169.72, 3180.16, 3405.90.

7.5 Nitroxyl

```
3
nitroxyl; constrained OO-DFT minimum; coordinates in Angstrom
O   0.0826117682  0.0000000000  1.1867248604
N  -0.1842703289 -0.0000000000 -0.0200741732
H   0.6016583607 -0.0000000000 -0.6666508772
```

Projected harmonic frequencies (cm^{-1}): 1261.41, 1592.58, 3301.05.

8 Timing interpretation

Earlier values labelled TDA were built with a symmetrized static response and omitted the nonlocal exchange difference required for the TDA A matrix. Their timing ratios and TDA accuracy claims are withdrawn. The archival measurements are explicitly identified in `data/archival_static_response_timing.json`; the corresponding CSV fields are labelled archival static response. Corrected TDA uses the corrected GFN2 ground orbitals, with effective exact exchange 0.225, and agrees with the independent PySCF matrix within 3.6×10^{-15} E_h.

The batched analytic Jacobian comparison uses the identical retained 44-AO nitropyridine checkpoint and two Oracle threads: 4.13837 versus 0.241612 s, maximum elementwise difference 8.88×10^{-16}. This 17.13-fold kernel speedup excludes reference preparation and does not establish an end-to-end advantage. The latest response/ROKS/relative-pair/trial-guard regression suite passes 31 tests on Oracle, including trusted Newton acceleration for resolved soft saddles.

The reduced extension initializes stationary parents from independently assigned NTO and native seeds. It applies local CSF overlap 0.90, attainable single-pair transition fidelity squared above one half, occupied/virtual norms 0.7 and an available-root projected margin of 0.05. The ratio to the leading singular value is a representability diagnostic, not a state probability. Absolute active-orbital anchor overlap is reported diagnostically when a fixed reference transition is supplied. Stationarity and curvature criteria remain unchanged. This assignment amendment is separated from the original 19-state data and from the subsequently locked accuracy assessment. The reduced scope is 14 states in ten molecules, selected after baseline convergence outcomes; no excluded state is silently removed from the original ledger.

9 Completed reduced extension

All 14 predictions and matched TDA assignments were locked before label opening. The scope was reduced after inspecting convergence and is not wholly blind. The original prediction lock, scoring output, label-opening record and full scope ledger accompany the document in `data/reduced_extension/`. Failed or unresolved earlier attempts remain in the calculation archive.

Table S5: Every reduced-extension energy (eV). The original prediction lock precedes label opening. TDA uses the corrected GFN2 ground orbitals.

ID / molecule	Family	TBE	OO	TDA-F	TDA-Q	TDA mean
B001 Diazirine	$n\pi^*$	4.093	4.242	5.247	4.230	4.739
B004 Nitropyridine_N-oxide	$n\pi^*$	4.395	4.263	4.096	4.515	4.305
B005 Phthalimide	$n\pi^*$	4.186	4.070	4.029	4.004	4.016
B007 Adenine	$n\pi^*$	5.304	4.805	4.245	5.192	4.719
B011 Aza-naphthalene	$n\pi^*$	5.346	5.279	5.054	5.507	5.281
B014 Adenine	$\pi\pi^*$	5.191	5.449	6.192	5.494	5.843
B015 Bimane_Syn	$\pi\pi^*$	3.964	3.911	5.189	4.947	5.068
B016 Cyclopentadiene	$\pi\pi^*$	6.451	8.197	8.282	6.624	7.453
B022 Furan	$\pi\pi^*$	6.530	7.974	8.448	6.829	7.638
B024 Phthalimide	$\pi\pi^*$	4.600	5.869	5.670	4.927	5.298
B025 Quinoxaline	CT	6.225	6.250	5.718	5.933	5.825
B029 Nitropyridine_N-oxide	CT	4.078	4.153	4.765	4.433	4.599
B030 Quinoxaline	CT	4.647	5.108	6.123	4.562	5.343
B039 Benzoquinone	Rydberg	7.083	7.681	8.582	7.212	7.897

10 Complete furan multi-CSF Hessian check

This check concerns the experimental full-relative-active-frame multi-CSF functional, separately from the original single-CSF minima and derivative checks. All 27 Cartesian columns are included; the maximum error is 2.2852×10^{-8} $E_\mathrm{h}/\mathrm{bohr}^2$. Electronic solutions are reoptimized at each displaced point, and all three ground-reference terms are included. Completed points and columns are retained for restart.

Table S6: Complete 27-column furan total excited-state Hessian validation. Analytic constrained orbital/CI response and all three ground-reference derivatives are included. Central differences reoptimize the state at each displaced geometry with step 2×10^{-4} bohr. Units are E_h/bohr for the gradient check and E_h/bohr^2 for the Hessian.

Cartesian column (zero based)	Energy–gradient error	Max. Hessian error
0	1.35e-15	1.25e-08
1	1.63e-09	1.65e-08
2	8.17e-09	1.52e-08
3	7.11e-11	1.25e-08
4	2.07e-09	1.65e-08
5	8.33e-09	1.52e-08

Table S6: Complete 27-column furan total excited-state Hessian validation. Analytic constrained orbital/CI response and all three ground-reference derivatives are included. Central differences reoptimize the state at each displaced geometry with step 2×10^{-4} bohr. Units are E_h/bohr for the gradient check and E_h/bohr^2 for the Hessian.

Cartesian column (zero based)	Energy–gradient error	Max. Hessian error
6	1.87e-10	9.93e-09
7	7.82e-09	1.83e-08
8	6.42e-09	2.28e-08
9	1.11e-10	9.93e-09
10	7.84e-09	1.83e-08
11	6.68e-09	2.29e-08
12	1.38e-10	1.89e-09
13	2.22e-11	8.80e-09
14	3.88e-09	2.80e-09
15	2.22e-10	2.13e-09
16	5.37e-09	1.16e-08
17	2.19e-10	2.98e-09
18	7.99e-11	2.13e-09
19	5.20e-09	1.16e-08
20	1.57e-10	2.98e-09
21	6.66e-11	1.56e-09
22	9.31e-10	2.77e-09
23	3.33e-09	9.41e-09
24	2.93e-10	1.56e-09
25	7.62e-10	2.78e-09
26	3.14e-09	9.41e-09

11 Reproducibility and automatic integration

The final remote qualification used `oracle.sns.it`. A transfer audit checked 561 result files and 30 source files; the remote test suite reports 86 passing tests. Cold and resumed executions of the integrated example differ by 1.7764×10^{-15} E_h. The copied `data/PROTOCOL.json` identifies the validated source files by SHA256. The source hashes refer to the calculation campaign; the manuscript-generation manifest separately identifies the document inputs and outputs.

The explicit interface selects the following settings:

```
native_excited_method = "roks_50_50"
native_roks_composite_orbital_coupling = "relative_active_pair"
native_roks_composite_following = "character_newton_index"
native_roks_composite_reference_policy = "geometry_only"
native_roks_composite_ground_surface = "native_plus_gaps"
native_roks_composite_component_tracking = "active_pair"
```

The example source is `examples/roks_relative_pair.py` in the MOUSE repository. Response paths, two certified seed paths, promotions, and parent saddle indices are supplied explicitly. The accepted generation has precedence over cold seeds. It stores both CSFs, relative targets, multipliers, three restricted ground references, model fingerprints, and the anchor. Defaults for other methods are not changed.

The manuscript-generation script `build_assets.py` performs lightweight data aggregation and plotting only. It can be run from the associated MOUSE working tree to regenerate all tables, the validation figure, optimized XYZ files, and TBE method provenance. The accompanying CSV provides all 19 OO-DFT values and explicitly labelled archival static-response controls. The completed reduced-extension corrected TDA and OO values are reported separately below. The four-molecule comparison in the main text is the designated prospective comparison; additional exposed TDA values in the CSV must retain that exposure label.

12 Public multi-CSF interface qualification

The explicit research interface is

```
native_excited_method = "multicsf_50_50"
native_multicsf_model_path = Path(".../model.json")
```

The model defines promotions, rank, reference transitions and the full relative frame. The example is `examples/multicsf_oo_dft.py`. Its input hashes are checked before optimizer-cache access. Accepted generations store both wavefunctions, CI coefficients, three ground references and derivatives; trial evaluations do not publish a new generation. A relocated complete model restores the same accepted state. Checkpoint corruption and modified model arrays are rejected. These checks and 46 distinct regression tests ran on Oracle with two threads. No electronic job ran on the laptop.

The cold full-Hessian interface call takes 98.99 s; restoring the accepted state and its already stored Hessian takes 0.613 s. The latter is not a fresh-Hessian timing. Surface evaluation is instrumented against SCC calls and invokes no DFT/TDA solver. Preparation still requires the archived DFT response and reference transition. All source/result hashes, native-extension hash, logs and portable furan inputs are recorded in the multi-CSF integration qualification archive. The original single-CSF automatic interface above remains a separate model.

The response-only λ diagnostic uses 0.85, 0.90 and 0.95, with the one-body DFT reference, semilocal contribution and bare off-diagonal CSF couplings held fixed. Vertical weight and fixed-reference rank perturbations are separate algebraic diagnostics; no new molecular minimum or weighted Hessian is inferred. The actual displaced-reference rank diagnostic is reported below; full F/Q energy recalibration and independent validation remain open.

13 Symmetry-assigned TD-DFTB and ΔDFTB comparison

All electronic calculations in this table were already completed on Oracle; the new audit only reads stored wavefunctions, transition weights, occupations and energies. DFTB2 uses mio-1-1 parameters. The auxiliary Gaussian s/p shells define the same angular group representation and are never used to compute an electronic energy or a cross-model radial overlap. Reference axes are reused explicitly to avoid irrep-name changes from equivalent axis conventions. A TD root requires more than 0.999 of the reported transition weight in its assigned irrep. The reference irrep is obtained from its AO transition density, allowing orbital degeneracy and mixing. The ΔDFTB final occupied spin orbitals must have irrep purity above 0.999; their product gives the broken-symmetry determinant's spatial irrep. Only globally first reference singlets are considered for this lowest-state method.

The 19-state TD-DFTB MAE is 0.54547 eV. For the 15 eligible completed ΔDFTB states, its MAE is 0.31353 eV and the same-subset MOUSE MAE is 0.17174 eV. On the four previous prospective cases, TD-DFTB gives 1.05835 eV MAE versus 0.30363 eV for MOUSE; on the three eligible prospective ΔDFTB cases, the respective MAEs are 0.23380 and 0.22655 eV. The original calculations, including unsuccessful SCC attempts for uracil and cyclopentadienone, are retained. No failed calculation is replaced by an assigned energy. Additional bounded Anderson

and linear-mixing recoveries retain the same mio-1-1 Hamiltonian, integer occupations and 10^{-10} SCC tolerance; both incomplete cases remain unresolved. The backend rejects binary charge restart for this method, so the final recoveries supply the previously converged atomic ground charges through the supported InitialCharges input. All 112 recovery artifacts, including input rejections and unsuccessful SCC histories, are retained separately from the 254 original audit inputs/results. Neither the group-theoretical audit nor the arithmetic score is a cross-model transition-overlap qualification or a new independent validation set.

Table S7: TD-DFTB2 and spin-purified ΔDFTB2 (mio-1-1) on the identical original 19-state geometries. TD assignments use a pure spatial irrep and the within-irrep ordinal of the fixed DFT reference. ΔDFTB is restricted to the first global reference singlet and requires the final broken-symmetry determinant to have the target spatial irrep. These are symmetry/ordinal comparisons without cross-method transition-overlap qualification. All energies are in eV. "SCC" denotes an incomplete archived calculation; "higher" denotes an inapplicable higher reference state.

Molecule	State	TBE	MOUSE	TD-DFTB	ΔDFTB
acetaldehyde	$n\pi^*$	4.304	4.277	4.531	4.766
acetone	$n\pi^*$	4.468	4.338	4.629	4.922
acrolein	$n\pi^*$	3.723	3.771	3.096	3.576
cyclopentadienone	$n\pi^*$	2.951	2.953	2.318	SCC
formamide	$n\pi^*$	5.644	5.648	5.651	6.002
glyoxal	$n\pi^*$	2.877	2.752	2.346	2.621
maleimide	$n\pi^*$	3.810	3.643	2.729	3.134
nitrosomethane	$n\pi^*$	1.952	1.721	2.068	2.658
nitroxyl	$n\pi^*$	1.743	1.485	1.729	2.336
pyranone	$n\pi^*$	4.016	4.076	3.436	4.012
pyrazine	$n\pi^*$	4.144	3.932	3.767	4.086
pyrimidine	$n\pi^*$	4.451	4.218	4.277	4.488
tetrazine	$n\pi^*$	2.462	2.060	2.387	2.713
uracil	$n\pi^*$	4.935	4.885	3.809	SCC
uracil	$\pi\pi^*$	5.381	5.686	4.980	higher
cyanoformaldehyde	$n\pi^*$	3.817	3.989	3.946	4.274
cyclopropenone	$n\pi^*$	4.274	4.482	3.342	4.147
imidazole	$n\pi^*$	6.715	6.180	9.569	higher
pyridazine	$n\pi^*$	3.830	3.530	3.511	3.947

14 Complete cyclopentadiene multi-CSF Hessian check

All 33 Cartesian columns pass through the public surface, with maximum Hessian error 1.2964×10^{-7} $E_\mathrm{h}/\mathrm{bohr}^2$ and maximum energy–gradient difference 9.0270×10^{-9} $E_\mathrm{h}/\mathrm{bohr}$. Accepted anchor and all 66 signed energy/gradient evaluations are retained. A separate lightweight reduction of the transferred checkpoints reproduces every reported error. The cold analytic Hessian call takes 250.27 s, with peak process RSS 878220 KiB; this timing begins after model loading and includes direct state relaxation and all three ground-reference derivative terms. The 66 displaced energy/gradient calls take 465.47 s in total, with median 7.05 s (range 4.07–11.27 s); thus the measured finite-difference Hessian cost is 1.86 times the analytic call for this prepared model. Both timings exclude DFT/model preparation and do not constitute a TDA comparison.

Both furan and cyclopentadiene stored Hessians are symmetric and satisfy the translational acoustic identity and nonstationary rotational-covariance identity $H(\Omega R) = \Omega g$. The largest translation residual is below 2×10^{-15} E_h/bohr^2, the largest rotation residual below 4×10^{-11} E_h/bohr, and the net torque below 3×10^{-15} E_h. These identities do not imply a molecular minimum or an experimentally accurate vibrational spectrum.

Table S8: Complete 33-column cyclopentadiene total-excited-energy Hessian check through the public multi-CSF surface. The fixed rank-two model includes native S0 restoration and both corrected ground subtractions. Each signed displacement is independently reoptimized from the accepted anchor; central step is 2×10^{-4} bohr. Units are E_h/bohr and E_h/bohr^2.

Cartesian column (zero based)	Energy–gradient error	Max. Hessian error
0	2.66e-11	7.48e-09
1	5.33e-11	4.07e-09
2	1.60e-09	1.20e-08
3	8.88e-11	1.16e-08
4	5.02e-09	1.89e-08
5	9.03e-09	1.92e-08
6	3.55e-11	1.16e-08
7	5.03e-09	1.89e-08
8	8.94e-09	1.92e-08
9	5.33e-11	1.30e-07
10	6.93e-09	1.58e-08
11	5.95e-09	2.52e-08
12	8.88e-11	1.30e-07
13	7.13e-09	1.58e-08
14	5.83e-09	2.52e-08
15	3.54e-09	6.84e-09
16	1.78e-10	1.52e-09
17	8.95e-10	4.41e-09
18	3.86e-09	6.83e-09
19	2.00e-10	1.52e-09
20	1.03e-09	4.42e-09
21	3.11e-11	3.04e-09
22	6.71e-09	1.88e-08
23	4.86e-10	2.95e-09
24	9.77e-11	3.03e-09
25	6.63e-09	1.88e-08
26	5.66e-10	2.95e-09
27	1.60e-10	5.83e-09
28	4.23e-10	2.99e-09
29	3.66e-09	1.01e-08
30	1.02e-10	5.83e-09
31	5.43e-10	2.99e-09
32	3.68e-09	1.01e-08

15 Actual DFT-reference rank and subspace test

The same three rank-two molecules are recomputed at uniform internal Cartesian scales 0.98 and 1.02 about the geometric center. The state-identity thresholds and nearest-integer effective-pair rule are unchanged. Two same-symmetry singlet roots must converge at residual tolerance 10^{-7} Hartree; independently recomputed full-operator residuals certify the final vectors. The final reference solver explicitly restricts the occupied/virtual pair space to the requested irrep, uses two-vector Davidson increments and linear-dependence threshold 10^{-20}, and saves positive-root Ritz checkpoints. Both previously completed neighboring roots can be transported by exact cross-AO overlap. Forbidden-sector numerical components from earlier full-space restart attempts are rejected; their logs and uncertified vectors remain archived and contribute no accepted result. Completed SCF checkpoints are reused. The six final points and baseline reprojection audit ran on Oracle with Python 3.11.15, NumPy 2.3.5 and PySCF 2.14.0; the native extension hash matches the original integration qualification.

The fresh-native-S0 and transported-original-subspace calculations both retain rank two. Baseline compact transition matrices are reproduced within 5×10^{-16}, before the changed references are analyzed. All six target transition overlaps exceed 0.996 and available-competitor margins exceed 0.972. This is a local reference/identity test with exposed data and limited competing-root coverage, not a full F/Q recalibration, a common-target-geometry energy anchoring test, or a new independent benchmark.

Table S9: Actual displaced-reference rank test. PBE0/aug-cc-pVDZ/grid-3 ground states and two same-symmetry TDA roots are recomputed at uniform Cartesian scales 0.98 and 1.02. A fresh restricted native GFN2 S0 defines the compact projection. N_{pair} and rank refer to that normalized compact transition. f is coherent cross-geometry transition overlap with the original fixed DFT root; p is the compact projection norm. This does not recalibrate the complete F/Q energy model.

Molecule	Scale	N_{pair}	Rank	f	p
Cyclopentadiene	0.98	1.9927	2	0.9982	0.8256
Cyclopentadiene	1.02	1.9850	2	0.9985	0.8511
Furan	0.98	2.0135	2	0.9984	0.8320
Furan	1.02	2.0071	2	0.9986	0.8585
Phthalimide	0.98	1.5924	2	0.9962	0.9318
Phthalimide	1.02	1.5982	2	0.9962	0.9317